behavior trees to tackle challenges such as code competitions and social media content creation.

- *Chapter 7, "Assembling and using an agent platform"* —This chapter introduces Nexus, a sophisticated platform for orchestrating multiple agents and LLMs. We discuss how Nexus facilitates agentic workflows and enables complex interactions between agents, providing an example of a fully functioning multi-agent environment.
- *Chapter 8, "Understanding agent memory and knowledge"* —Retrieval-augmented generation (RAG) has become an essential tool for extending the capabilities of LLM agents. This chapter explores how retrieval mechanisms can serve as both a source of knowledge by processing ingested files, and of memory, allowing agents to recall previous interactions or events.
- *Chapter 9, "Mastering agent prompts with prompt flow"* —Prompt engineering is central to an agent's success. This chapter introduces prompt flow, a tool from Microsoft that helps automate the testing and evaluation of prompts, enabling more robust and effective agentic behavior.
- *Chapter 10, "Agent reasoning and evaluation "*—Reasoning is crucial to solving problems intelligently. In this chapter, we explore various reasoning techniques, such as chain of thought (CoT), and show how agents can evaluate reasoning strategies even during inference, improving their capacity to solve problems autonomously.
- *Chapter 11, "Agent planning and feedback"* —Planning is perhaps an agent's most critical skill in achieving its goals. We discuss how agents can incorporate planning to navigate complex tasks and how feedback loops can be used to refine those plans. The chapter concludes by integrating all the key components—actions, memory and knowledge, reasoning, evaluation, planning, and feedback—into practical examples of agentic systems that solve real-world problems.

About the code

The code for this book is spread across several open source projects, many of which are hosted by me or by other organizations in GitHub